1	20	
2	25	
3	10	
4	15	
5	15	
6	15	
Total	100	

Probability theory I, Fall 2023

Quiz 2

November 21, 2023

Name:______________________________

Student No.:______________________________

1. Are the following random variables discrete or continuous? Give the probability mass function or density function based on your argument.

   (a) (10 points) X is an exponential random variable with parameter $\lambda > 0$, let $Y = \lfloor X \rfloor$. Recall that $\lfloor x \rfloor = \sup\{n \in \mathbb{N} : n \leq x\}$

   (b) (10 points) W is a normal random variable with parameters $\mu = 0, \sigma^2 = 4$, let $V = W^2$.

2. Answer the following questions and briefly give the reasons (5 points for each):

   (a) X follows the uniform distribution on $(0, 10)$, find its hazard rate function.

   (b) Tossing a fair die N times, and N follows Poisson distribution with parameter 30, give the distribution of the appearance times of 6.

   (c) The distribution function of X is $F(x) = \begin{cases} \frac{1}{2} + \frac{1}{\pi} \arctan x, & x \leq 1 \\ 1, & x > 1 \end{cases}$, is X a continuous random variable?

   (d) X, Y are independent and follow the exponential distribution with parameter 1, is $X + Y$ an exponential random variable?

   (e) Z follows standard normal distribution, are Z, Z^2 independent?

3. (10 points) X is a random variable taking values on $[0, M]$, prove that $\mathrm{Var}(X) \leq \frac{M^2}{4}$.

4. 720 independent rolls of a fair die will be made. Use the normal approximation to estimate the following probabilities (write in the forms that you can check the distribution table for χ).

   (a) (7 points) Estimate the probability that the number 4 will appear between 100 and 135 times **inclusively**.

   (b) (8 points) If the number 4 appears exactly 120 times, estimate the probability that the number that is greater than(>) 4 will appear more than(>) 215 times.

5. The joint probability density function of X, Y is $f(x, y) = c(y^2 - x^2),\ -y \leq x \leq y, 0 < y < 1$.

   (a) (5 points) Find the value of c;

   (b) (10 points) Find $\mathbb{E}[X]$.

6. X, Y are independent random variables, and follow exponential distributions with respective parameters λ_1, λ_2.

   (a) (10 points) Find the probability density function of X/Y.

   (b) (5 points) Compute $\mathbb{P}(X < Y)$.

scratch paper